

REQUIREMENT ANALYSIS – FLIPKART E-COMMERCE DATABASE SYSTEM

1. Understanding the Flipkart E-Commerce Business Process

Flipkart is an online e-commerce marketplace where customers can search for products, compare prices, add items to a cart, place orders, make payments, and track deliveries. Sellers use the platform to list products, manage inventory, receive orders, and fulfil customer purchases. A database is required to connect these activities and maintain consistent information.

1. Customer registration/login – The customer creates an account or signs in.
2. Product discovery – The customer searches, filters, compares, and views product information.
3. Cart and wishlist management – Products can be added, removed, or saved for later.
4. Order placement – The customer confirms products, quantities, delivery address, and order details.
5. Payment – The system records the selected payment method, amount, transaction reference, and status.
6. Order fulfilment – The seller receives and processes the order.
7. Shipping and delivery – The order is packed, dispatched, tracked, and delivered.
8. After-sales service – The customer may request a return/refund, contact support, or submit a product review.

2. Data Required for Customer Management

For a Flipkart-style system, customer management requires information that supports account access, communication, delivery, orders, and customer service.

Data Field	Purpose	Example
Customer ID	Unique customer identification	C10001
Full Name	Customer identification	Ravi Kumar
Email Address	Login and communication	ravi@example.com
Phone Number	Contact and delivery communication	9876543210
Password	Account authentication; stored securely	Hashed value
Delivery Address	Product delivery	Chennai, Tamil Nadu
Billing Address	Billing and invoices	Chennai, Tamil Nadu
Registration Date	Account tracking	28-08-2026
Order History	Customer purchase tracking	ORD1001, ORD1002

3. Major Entities in the Flipkart E-Commerce Database

Customer: Stores customer account, contact, address, and profile information.

Seller: Stores seller details and marketplace-related information.

Product: Stores product name, description, price, category, stock, and seller reference.

Category: Groups products into categories.

Cart: Stores products selected by a customer before checkout.

Order: Stores an order placed by a customer, including date, amount, address, and status.

Order Item: Stores each product and quantity contained in an order.

Payment: Stores payment amount, method, transaction reference, and payment status.

Shipment: Stores dispatch, tracking, carrier, and delivery status.

Review: Stores customer ratings and reviews for products.

4. Key Relationships

- A Customer can place many Orders; each Order belongs to one Customer.
- A Seller can list many Products; each Product is associated with a Seller.
- A Category can contain many Products; each Product belongs to a category in this simplified design.
- An Order contains one or more Order Items.
- Each Order Item refers to one Product; a Product can appear in many Order Items.
- An Order has a Payment record containing transaction information.
- An Order can have a Shipment record used to track delivery.
- A Customer can write Reviews for products they have purchased.

5. ER Diagram – Flipkart E-Commerce Database

Simplified ER representation (PK = Primary Key, FK = Foreign Key):

CUSTOMER (Customer_ID PK, Name, Email, Phone, Password, Address)

1 CUSTOMER places N ORDER

ORDER (Order_ID PK, Customer_ID FK, Order_Date, Total_Amount, Status)

1 ORDER contains N ORDER_ITEM

ORDER_ITEM (Order_Item_ID PK, Order_ID FK, Product_ID FK, Quantity, Unit_Price)

N ORDER_ITEM refers to 1 PRODUCT

PRODUCT (Product_ID PK, Seller_ID FK, Category_ID FK, Name, Price, Stock)

N SELLER listed by 1 PRODUCT

SELLER (Seller_ID PK, Seller_Name, Email, Phone)

N SELLER belongs to 1 CATEGORY

CATEGORY (Category_ID PK, Category_Name)

ORDER (Order_ID PK) 1 ORDER has 1 PAYMENT

PAYMENT (Payment_ID PK, Order_ID FK, Amount, Method, Transaction_Ref, Status)

ORDER (Order_ID PK) 1 ORDER tracked by 1 SHIPMENT

SHIPMENT (Shipment_ID PK, Order_ID FK, Tracking_ID, Status, Delivery_Date)

CUSTOMER 1 CUSTOMER writes N REVIEW N CUSTOMER about 1 PRODUCT

REVIEW (Review_ID PK, Customer_ID FK, Product_ID FK, Rating, Comment)

Cardinality summary: Customer 1:N Order; Seller 1:N Product; Category 1:N Product; Order 1:N Order Item; Product 1:N Order Item; Order 1:1 Payment; Order 1:1 Shipment; Customer 1:N Review; Product 1:N Review.

6. Review Questions and Answers

Q1. Why does an e-commerce system require a database?

A Flipkart-style e-commerce system handles a large amount of information, including customers, sellers, products, prices, stock, carts, orders, payments, shipments, and reviews. A database stores this information in an organized way, allows fast retrieval and updates, reduces duplication, maintains consistency, supports transactions, and helps the platform generate reports and provide reliable services.

Q2. What customer details need to be stored?

The system should store Customer ID, full name, email address, phone number, securely stored password information, delivery addresses, billing information where required, registration date, and order history. Additional information such as saved addresses, preferences, support records, and reviews may also be maintained according to business requirements.

Q3. What operations should be performed on customer data?

- Create – Register a new customer account.
- Read – Retrieve customer profile and required account information.
- Update – Modify permitted details such as name, phone number, email, password, or address.
- Delete/Deactivate – Delete or deactivate an account according to applicable business and legal requirements.
- Search – Find customers using permitted fields such as Customer ID, email, or phone number.
- Validate – Check required fields, formats, and duplicate account conditions.
- Secure – Apply authentication, authorization, encryption/hashing, and access controls to protect customer data.

7. Conclusion

A Flipkart-based e-commerce database must support the complete marketplace process from customer registration and product discovery to ordering, payment, shipment, delivery, reviews, and after-sales service. Proper identification of entities, attributes, relationships, and customer-data operations provides a strong foundation for an efficient and reliable e-commerce system.